

House Walls (No Roof)

A loop inside a loop

반복 안의 반복으로 벽 만들기

BIG PICTURE

From a line to a wall



One loop makes a line. A loop *inside* a loop makes a wall.

한 줄이 모이면 벽이 돼요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

A wall has length **and** height

Long loop builds across. Tall loop builds up.

```
repeat 10 times:  
  place block down  
  move forward
```

This is a flat wall **10 long** on the ground.

길이 + 높이 = 벽

CONCEPT

Nested loop = a loop inside a loop

```
repeat 10 times:      # 10 columns
  repeat 10 times:    # 10 tall each
    place block down
    move up by 1
  move forward
```

Inner loop = one tall column. Outer loop = move along 10 times.

안쪽 반복이 기둥, 바깥 반복이 길이

CONCEPT

How many blocks?

10 long
× 1 tall = 10

10 tall
× 1 wide = 10

10 × 10
= 100 blocks

nested
multiplies!

10 × 10 = 100개

CODE

Blocks: a 10×10 wall

```
repeat 10 times:  
  repeat 10 times:  
    place block down  
    move up by 1  
  move forward
```

100 blocks — built from just 5 lines.

CODE

The same wall in Python

```
for col in range(10):  
    for row in range(10):  
        agent.place(DOWN)  
        agent.move(UP, 1)  
        agent.move(FORWARD, 1)
```

A teacher tool uses `blocks.fill` to do this in one line:

```
blocks.fill(OAK_PLANKS, pos(10,0,0), pos(10,3,6))
```

중첩 반복 = nested loop

PREDICT 🤖

What shape? How many blocks?

Predict first — then click reveal

```
repeat 10 times:  
  repeat 10 times:  
    place block down  
    move up by 1  
  move forward
```

Reveal output 🔍 reveal

VOCAB

New words today

nested loop a loop placed inside another loop

wall blocks placed long and tall

column a single tall stack of blocks

height how many blocks tall something is

중첩 반복

벽

기둥

높이



DEBUG

What's wrong with this code?

Goal: a wall **10 blocks long** on the ground.

buggy

```
repeat 9 times:  
  place block down  
  move forward
```

Think first — then click next.



DEBUG

The bug

Bug: `repeat 9` only places **9** blocks. An off-by-one mistake — for a 10-long wall you need `repeat 10`.

fixed

```
repeat 10 times:  
  place block down  
  move forward
```

YOUR TURN!

Build house walls

✓ Open your homework world + Code Builder

✓ Use **two loops** (nested) for one wall

✓ Make it at least **10 long and 10 tall**

✓ No roof yet — just standing walls

✓ Send a photo of your walls



QUIZ

QUESTION 1 OF 3

What is a **nested loop**?

A loop placed inside another loop.

Two commands on one line.

A loop with no count.

A loop that never stops.



QUIZ

QUESTION 2 OF 3

A 10×10 wall is how many blocks?

20

10

100

1000



QUIZ

QUESTION 3 OF 3

To place exactly **10** blocks in a line you write:

repeat 11 times

repeat 10 times

repeat 1 time

repeat 9 times

QUIZ

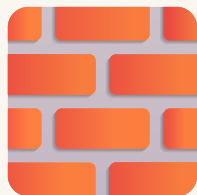
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Walls up!

Next: add a roof with variables.

다음 주: 변수로 지붕 올리기